ENTERPRISE STRATEGY & INNOVATION

Drivers of Digital Transformation

Understanding the Market, Technology, and Customer Forces Shaping the Future of Business

Singapore & Global Enterprise Context

The Triad of Transformation Drivers

1. Market Drivers

WHY CHANGE?

External business pressures, competitive threats, and economic shifts forcing organizations to rethink their strategy.

- Rising competitive pressure
- Globalisation & trade
- Disruption & government pushes

2. Technology Drivers

WHAT ENABLES CHANGE?

Cutting-edge digital tools and modern cloud infrastructure that create powerful new technical capabilities.

- Artificial Intelligence & GenAI
- Cloud Computing & IoT
- Digital Payments & APIs

3. Customer Drivers

WHAT IS EXPECTED?

Evolving customer behaviors, preferences, and demand for instant, frictionless digital experiences.

- 24/7 convenience & speed
- Hyper-personalisation
- Omnichannel interactions

Market Drivers: Pressures to Adapt

Increasing Competition: Local firms face fierce competition from digital-first platforms (e.g., Grab, Shopee, Lazada) and global digital giants.

Digital Economy Growth: Singapore's digital economy accounted for **18.6% of GDP in 2024**, making digital participation essential.

Cost & Productivity Pressures: High operational costs drive businesses toward automation, AI, and streamlined workflows.

Government Initiatives: IMDA & EnterpriseSG provide roadmaps & grants (e.g., Refreshed Retail Industry Digital Plan 2026 guiding 2,000+ SMEs).

Technology Drivers: Enabler of Change

- **Artificial Intelligence & Generative AI:** Automates decision-making, streamlines knowledge work, improves service productivity, and powers customer chatbots.
- **Cloud Computing & APIs:** Replaces on-premise infrastructure with scalable platforms, connecting systems with banks, logistics, and marketplaces.
- **Data Analytics & IoT Sensors:** Real-time data streams monitor inventory and machine health, enabling precise forecasting and predictive operations.
- **Mobile Tech & Digital Payments:** Smartphones and payment rails (PayNow, e-wallets) support instant, frictionless, cashless business models.
- **Cybersecurity Infrastructure:** Protects digital assets and builds trusted ecosystems as operations transition fully online.

Spotlight: AI Adoption Acceleration

14.5%

SME AI ADOPTION (2024)

Up from 4.2% in 2023

62.5%

NON-SME ADOPTION

Up from 44.0% in 2023

The AI Value Chain

Transformation Progression:

AI Adoption → Automation → Better Productivity → New Digital Services → Novel Business Models

National AI Impact Programme: Targeted to support **10,000 enterprises** over 3 years to advance real-world AI deployment across Singapore.

Customer Drivers: Evolving Demands

Convenience & Speed

24/7 Availability: Expectation to shop, order, and access services anytime via smartphones.

Instant Fulfillment: Demand for immediate answers via AI chatbots and real-time payment settlement.

Cashless Convenience: Ubiquitous adoption of PayNow, mobile wallets, and contactless cards.

Experience & Choice

Hyper-Personalisation: Expectation that platforms remember purchase history and tailor recommendations.

Omnichannel Flexibility: Seamless movement between online research, store visits, and app purchases.

Price Transparency: Ability to instantly compare prices across regional platforms.

Case Study: FairPrice Group

OMNICHANNEL ECOSYSTEM TRANSFORMATION

Store of Tomorrow Programme

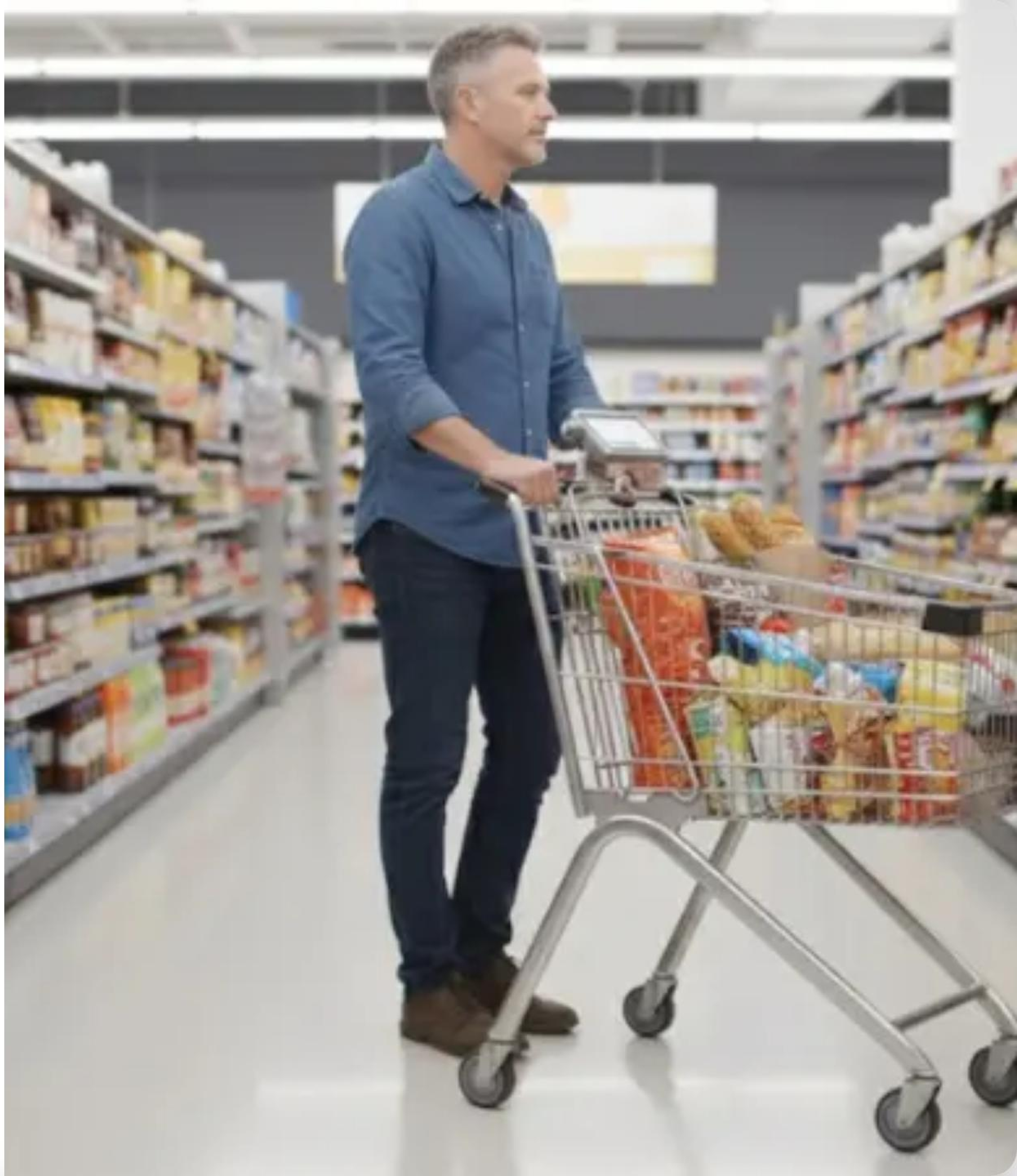
FairPrice transitioned from a traditional supermarket chain to an integrated, data-driven omnichannel retail ecosystem.

Market Driver: Intense competition from online grocery apps & productivity need.

Tech Driver: AI, smart carts, analytics, digital payments, mobile apps.

Customer Driver: Need for online shopping, seamless rewards, and speed.

Physical stores now incorporate smart carts and digital tools that sync directly with the mobile app and loyalty program.



Multichannel vs. Omnichannel

Dimension	Multichannel Strategy	Omnichannel Strategy
Channel Connection	Operates in silos (Store vs. Web vs. App)	Fully connected; real-time data sharing
Customer Data	Repeated information across channels	Data follows the customer seamlessly
Inter-Channel Dynamics	Channels operate independently or compete	Channels collaborate to complete transactions
Strategic Focus	Maximising sales touchpoints ("More ways to sell")	Unified customer experience ("One seamless journey")

The Connected Customer Journey

1. App Search

Customer browses items & adds them to a digital wishlist.

2. Physical Visit

Visits store to inspect items using smart cart guidance.

3. Digital Pay

Checkout via PayNow or linked mobile app wallet.

4. Delivery Choice

Opt for store pick-up or home delivery fulfillment.

5. Loyalty Sync

Earns Linkpoints & receives personalized digital receipts.

Business Disruption: Dropshipping

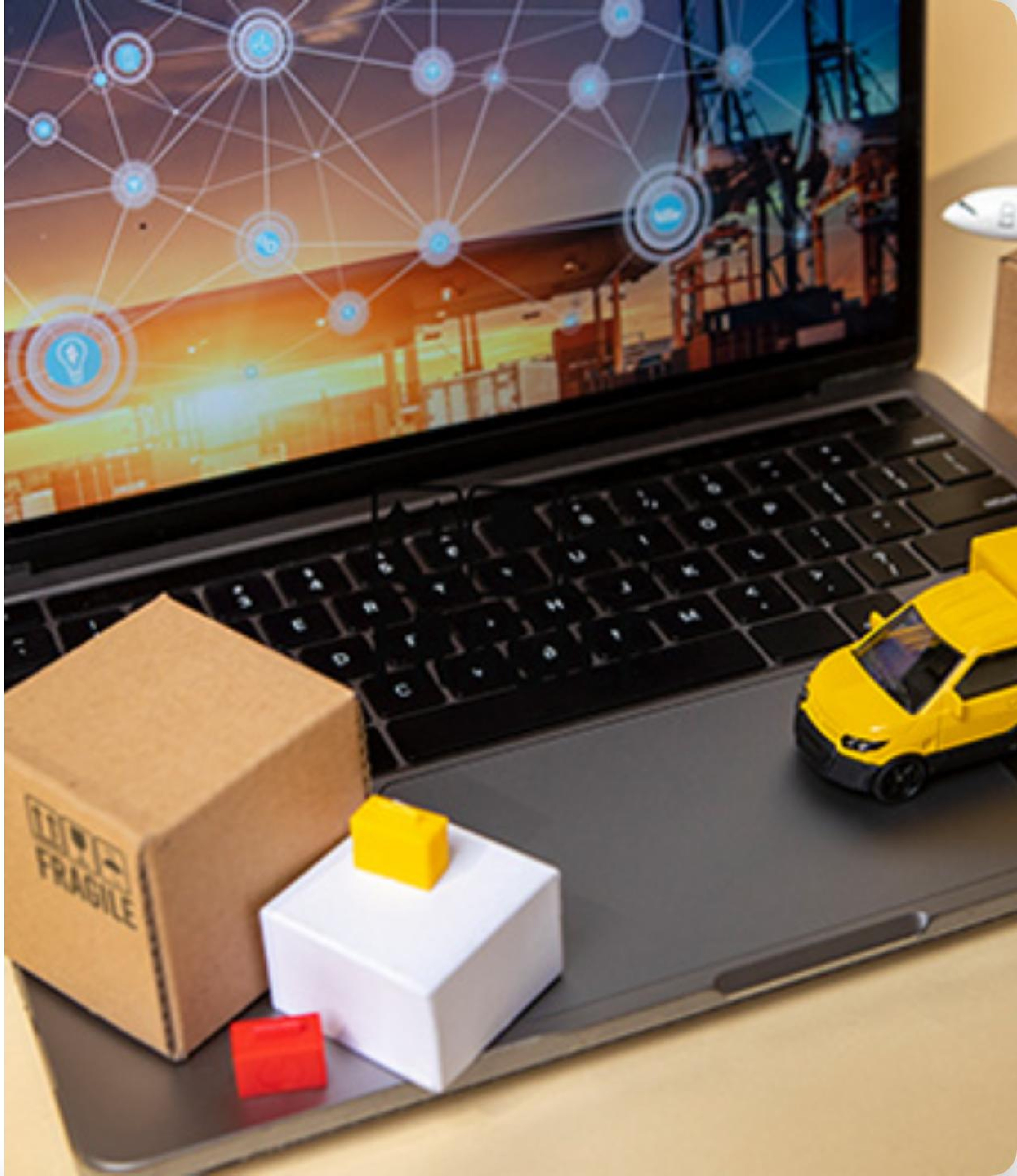
VALUE CHAIN RE-ENGINEERING

Separating Sales from Inventory

An e-commerce model where the seller does not store or physically handle inventory. Digital platforms connect buyers directly to suppliers.

How It Works:

1. Customer orders on store ($50) → 2. Store pays Supplier ($30) → 3. Supplier ships directly to Customer → 4. Seller keeps margin ($20).



Dropshipping vs. Traditional Retail

Operational Aspect	Traditional E-Commerce	Dropshipping Model
Inventory Ownership	Upfront bulk purchase required	No inventory ownership needed
Warehousing & Packing	Own/rented warehouse space	Supplier manages storage & packing
Capital Investment	High initial capital commitment	Low initial financial barrier
Primary Business Focus	Logistics, storage, inventory control	Digital marketing, UX, customer support

Questions & Discussion

Driving sustainable digital transformation through market, technology, and customer alignment.

LO3 Digital Transformation Series | Enterprise Education

Image Sources

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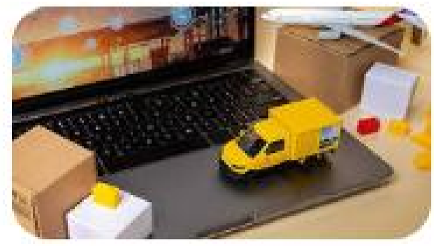

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